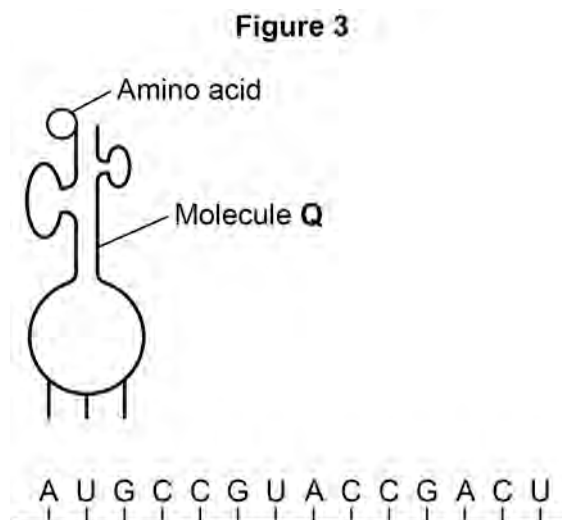


3 **Figure 3** represents one process that occurs during protein synthesis.



0 3 . **1** Name the process shown.

[1 mark]

0 3 . **2** Identify the molecule labelled **Q**.

[1 mark]

0 3 . **3** In **Figure 3**, the first codon is AUG. Give the base sequence of:

[2 marks]

the complementary DNA base sequence _____

the missing anticodon _____

Table 1 shows the base triplets that code for two amino acids.

Table 1

Amino acid	Encoding base triplet
Aspartic acid	GAC, GAU
Proline	CCA, CCG, CCC, CCU

- 0 3 . 4** Aspartic acid and proline are both amino acids. Describe how two amino acids differ from one another. You may use a diagram to help your description.

[1 mark]

- 0 3 . 5** Deletion of the sixth base (G) in the sequence shown in **Figure 3** would change the nature of the protein produced but substitution of the same base would not. Use the information in **Table 1** and your own knowledge to explain why.

[3 marks]

[Extra space]
